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Experimental study on the influence of particle size and type on single particle breakage of construction waste

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ABSTRACT

Construction waste is characterized by heterogeneous type, irregular morphology, and complex internal architecture. The primary constituents include recycled mortar (RM), recycled brick (RB), and recycled concrete (RC). To systematically investigate the comprehensive influence of different particle types and particle sizes on the breakage behavior of construction waste particles, a series of single-particle compression tests was conducted. The characteristic breakage strength and strength variability were quantitatively evaluated using the Weibull distribution function. Fractal dimension analysis was performed by comparing gradation curves before and after breakage. According to the experiments, the particle size and type have a coupled effect on single particle breakage behaviors of construction waste. The Weibull modulus (m) and fractal dimensions (D) show that with the increase of particle size, the breakage strength dispersion of RM and RB decreases, whereas that of RC increases. Particles with the lower breakage strength exhibited a greater propensity for fragmentation, demonstrating pronounced fractal characteristics consistent with established construction waste fractal models. Based on the force-displacement characteristics and fragmentation patterns, three distinct breakage modes were identified: intermediate splitting, mixed breakage, and bulk breakage. The proportions of different breakage modes varied with particle types and sizes. In general, as particle size increases, the dominant breakage mode of RM and RB transitions from bulk breakage to mixed breakage, whereas intermediate splitting mode becomes the predominant breakage mode of RC particles. The research results can provide technical insights into the breakage characteristics of construction waste materials and offer potential guidance for engineering applications.

1. Introduction

Accelerated urbanization in China has driven an annual increase in construction waste generation, with projections indicating a volume will surpass 4 billion metric tons by 2026. The national circular economy development strategy outlined in the "14th Five-Year Plan'' sets a target of 60 % recycling efficiency by 2025. Conventional disposal methods such as landfilling, open-air stockpiling, and uncontrolled incineration—are being gradually replaced by advanced recycling processes involving classification, crushing, and screening. These sustainable techniques produce recycled fillers that mitigate land scarcity, environmental degradation, and depletion

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of natural aggregates [19,35,56,60]. Recycled aggregates from construction waste primarily comprise recycled mortar (RM), recycled brick (RB), and recycled concrete (RC) [11,44,47,48], which together account for 70–80 % of total construction waste [24,36,52]. These recycled materials have been applied in subgrade compaction [32,46,55,57], soft-soil stabilization [10,59], ecological slope reinforcement [33], aggregate improvement [1,7,43]. In recent years, the recycling and reuse of construction and demolition waste (C&D) have attracted growing attention for promoting sustainable development. Recent studies have explored intelligent and low-carbon approaches for recycled material design, including hybrid machine learning and multi-objective optimization for green concrete [34], data-driven prediction of recycled aggregate concrete properties [23], and the synergistic use of recycled aggregates with waste glass powder [37]. These advances provide valuable insights for improving the performance and sustainability of recycled materials. However, recycled construction aggregates exhibit distinct characteristics compared to natural aggregates, including heterogeneous type, wide particle-size variability, and irregular morphology, making them more susceptible to particle breakage under external loading. Particle crushing alters the gradation characteristics of recycled aggregates [28], consequently modifying their macroscopic mechanical behavior [4,15,29].

Particle breakage mechanisms are governed by multifactorial interactions including loading regimes, parent material strength, particle geometry, and size distribution [14,20], which is generally investigated through single-particle compression test. [26]) established a micromechanical framework demonstrating that the bulk yield stress depends on the weakest-link principle, and the system strength is governed by the tensile resistance of the smallest constituent particles. Through integrated one-dimensional compression testing and single-particle loading analysis, Nakata et al. [29] elucidated the particle-scale determinants of macroscopic one-dimensional compression behaviors of sand. The study identified five key governing parameters: particle size distribution, initial void ratio, angularity characteristics, mineralogical type, and tensile strength, collectively controlling the compressibility of particle assembly. Shen et al. [38] analyzed the influence of particle size, shape, and other factors on the breakage strength and breakage mode of calcareous sand, and found that the characteristic strength of massive particles was the highest, and proposed three splitting modes and three breakage modes. Meng et al. [27] employed coupled experimental-computational approaches to investigate size-dependent breakage behavior of mudstone and sandstone particles. Based on the particle-size distribution before and after breakage, different materials exhibited distinct fractal dimensions and Weibull modulus, and the single-particle breakage strength showed a clear size effect. Yin et al. [53] carried out a single-particle compression test of course calcareous sand particles, and the results showed that the breakage strength decreased with the increase of particle size. Wu et al. [51] performed morphology-dependent fragmentation analyses on 200 coral sand particles, establishing four distinct failure progression patterns. According to the force-displacement curve obtained, four breakage modes-particle explosion, particle cracking, corner wear and particle breakage, were summarized. Current research paradigms on particle breakage mainly focus on brittle geomaterials such as calcareous sands and coral aggregates, with particular focus on size-shape-strength interrelationships, particle size scaling effects, and fragmentation pattern evolution. However, due to the fundamental differences in type and particle size between natural soils and construction waste aggregates, the breakage behaviors of construction waste particles are extremely complex, involving the interaction of multiple factors that have coupled influence on their breakage mechanisms and patterns.

In order to better understand the mechanical properties of construction waste, some scholars have carried out research on particle breakage behavior of recycled material. Xu [50] employed controlled blasting techniques on recycled construction waste, derived particle-size distribution curves through sieve analysis and the results revealed that the fractal dimensions (D) ranging from 2.50 to 2.60. Arulrajah et al. [2] quantified D values for four kinds of construction wastes: recycled concrete aggregate (RCA) at 2.61, recycled

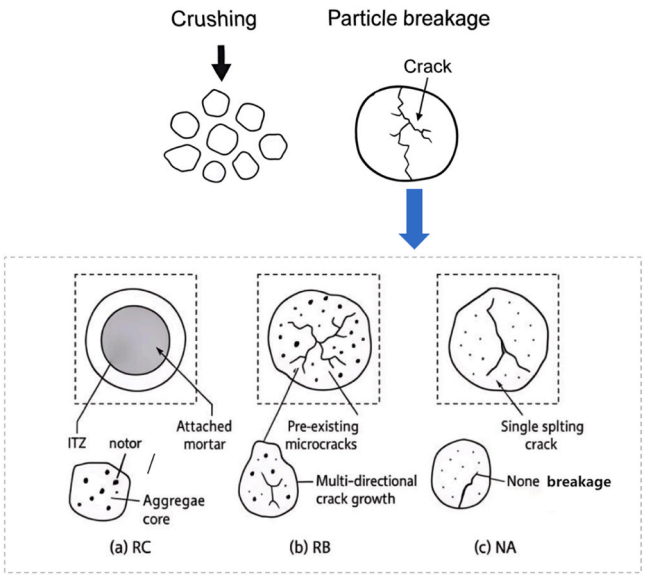


Fig. 1. Typical particle breakage mechanisms of recycled and natural aggregates.

asphalt pavement (RAP) at 2.52, crushed brick (CB) at 2.61, and waste excavation rock (WR) at 2.59, confirming the applicability of fractal theory on construction waste breakage processes. Oskooei et al. [30] through controlled single-particle compression tests that blocky particles (9.5–13.2 mm) exhibited maximum breakage resistance, corresponding to the D value of 2.588. Systematic analysis revealed an inverse correlation between Weibull modulus (m=2.0–2.2) and particle size, indicating enhanced strength heterogeneity and pronounced size dependence under controlled loading conditions. Oskooei et al. [31] studied the effect of particle shape on the breakage behavior of 4.75 mm to 19 mm recycled concrete aggregate based on a discrete element model, founding that Weibull modulus decreased with the increase of strength variability. Zhang et al. [58] evaluated the breakage characteristics of three construction waste materials (gravel, brick and mortar) with a particle size of 25 mm. According to the results, the single particle breakage strength of gravel was the strongest, while mortar was the weakest. Cai et al. [5] studied the particle shape and breakage characteristics of 19–37.5 mm RB, RC and RM particles, and proposed three breakage models- penetrating fracture, corner fracture and overall fragmentation-nothing that increase in sphericity promotes overall fracture. Park et al. [33] conducted a single-particle compression test on 170 waste concrete aggregates with particle sizes ranging from 30 mm to 40 mm, and identified a three-stage breakage process, namely surface breakage, mortar breakage and natural aggregate breakage.

In this study, "particle breakage" refers to the fracture or fragmentation of a single particle under external loading, which is controlled by intrinsic defects, pore structures, and the heterogeneity of internal stress distribution. In contrast, "crushing" describes the collective comminution process of multiple particles involving particle–particle interactions and contact stress transfer at the bulk scale [8,17,18]. RC and RB aggregates show distinct particle breakage features compared with natural aggregates (NA). RC particles contain attached mortar and weak interfacial transition zones (ITZs), leading to progressive crack propagation and layered fragmentation. In contrast, RB particles exhibit porous microstructures and pre-existing cracks, resulting in extensive multi-directional breakage and lower breakage resistance. Natural aggregates, by comparison, present denser structures and primarily exhibit single splitting failure modes (Fig. 1).

According to previous research, the primary constituent materials of construction waste (RC, RB, RM) exhibit distinct breakage behaviors, while particle size also significantly influences the breakage strength of construction waste particles. Recent studies have advanced the understanding of the single-particle breakage behaviour of construction and demolition waste (CDW) aggregates. Gao [12] reported that the breakage strength of recycled concrete and brick aggregates follows a Weibull distribution with a clear size dependence. Yosri et al. [54] highlighted that particle breakage of CDW under varying compaction and moisture conditions strongly influences their macroscopic strength and stiffness. Moreover, John et al. [16] confirmed that the synergistic interaction of CDW types



(a) RM



(b) RB



(c) RC

Fig. 2. Construction waste particles.

leads to distinct breakage characteristics governed by particle size and type. However, little attention has been paid to the coupled effects of these factors on single-particle breakage behaviors of construction waste. Therefore, the major objective of this study is to investigate the coupled effects of particle size and material type on breakage strength and breakage mode of construction waste particulates. Single-particle compression tests were conducted using a servo-controlled universal testing machine (UTM), with breakage strength determined from peak load values in force-displacement profiles. The Weibull statistical framework was employed to derive characteristic strength parameters, with reliability analysis conducted using survival probability curves to quantify strength variability across experimental conditions. Gradation analysis before and after fragmentation was performed using standardized sieve protocols, with fractal dimensions calculated via fractal theory to quantify comminution intensity. The results summarize the variations in breakage strength and breakage modes among different types and sizes of construction waste particles, providing insights into their coupled effects on single-particle breakage behavior.

2. Single particle compression experiment

2.1. Experimental material

The experimental materials consisted construction and demolition waste (CDW) aggregates sourced from Hangzhou, Zhejiang Province, China. Following manual sorting and mechanical processing, three dominant constituents were identified: RC, RB and RM. Particle morphology was standardized using the Sneed-Folk classification system [39], with block-shaped particles (axial ratio$\leq$1.5) selected based through quantitative shape analysis. The particle size range was determined with reference to Test Methods of Aggregates for Highway Engineering (JTG 3432–2024) and Recycled Coarse Aggregate for Concrete (GB/T 25177–2010). Considering experimental operability and uniform loading, all particles were fractionated into three size ranges (4.75–10 mm, 10–20 mm, 20–25 mm) using ASTM-standard sieves. thirty samples were tested for each particle size group for RC and RB. For RM, thirty-six specimens were prepared to account for loading instability, from which thirty valid samples were finally used for data analysis. In total 270 dry test specimens (30 per material per size fraction) as illustrated in Fig. 2. This range ensures test stability and data comparability. The physical properties of these materials are listed in Table 1.

2.2. Test instrument and scheme

An electronic universal testing machine (Fig. 3) equipped in the laboratory was used to carry out the single-particle compression test. A displacement-controlled loading was applied at 1 mm/min loading rate with 50 kN capacity load cell (ISO 7500–1 Class 0.5 certified), maintaining a force measurement uncertainty with $\pm$0.8 % throughout the tests. Three critical experimental challenges were identified: (1) Stress concentration anomalies due to morphological heterogeneity; (2) Premature fracture initiation caused by non-uniform stress distribution;(3) Boundary condition artifacts from improper fixturing. To minimize these issues, the shortest axis of each particle was first placed perpendicular to the loading platen to ensure stable positioning. Then, the loading beam was manually adjustied via remote control until the upper platen was fully in contact with the particle surface. Based on previous studies, the loading speed was set at 1 mm/min [5,30,56] to ensure uniform load application and high testing accuracy. For small particle size (4.75–10 mm), a metal pad was positioned between the two loading plates and the particles were centred on the pad for loading. A camera was used to record the entire breakage process and capture the morphological changes before and after fracture. Finally, the test was terminated when the force-displacement curve exhibited a steep drop accompanied by an audible cracking sound, or when the load no longer increased with displacement. At this point, visible major fractures were observed, and the bearing capacity had fallen to less than approximately 10 % of its peak value, indicating that the particle had reached its breakage strength.

2.3. Research framework and theoretical basis

The maximum tensile stress can be used to calculate the breakage strength of a single particle. When the construction waste particles are under vertical load, tensile stress will be occurred inside the particles, especially when the internal tensile stress exceeds the limit of the particles themselves, resulting in the particles breaking in a certain direction. Building upon the Hertzian contact theory framework, the characteristic breakage strength can be determined using the Hiramatsu-Oka equation [13], later refined by Wang et al. [41]:

$$\sigma_f = \frac{0.9F_f}{d_0^2} \tag{1}$$

Table 1
Physical properties of the materials.

physical index	Apparent densityρ_a (g/cm3)	Bulk densityρ_b (g/cm3)	Specific gravity G_S
RM	2.028	1.113	2.028
RC	2.217	1.296	2.217
RB	2.140	1.063	2.140

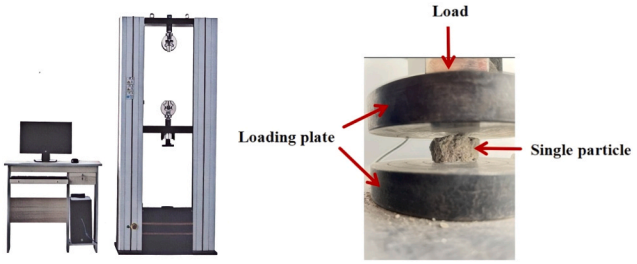


Fig. 3. Universal pressure testing machine.

in which $F_f$is the vertical force at the failure point; The equivalent particle size d_0 can be determined by measuring the long axis (L), central axis (I) and short axis (S) of the particle with a vernier caliper. There are five methods for measuring the equivalent particle size of irregular particles: For elongated and flaky particles, the long axis is much larger than the short and central axes, and the long axis of the particles has a small influence on particle breakage, d_0 can be calculated as $d_0 = \sqrt{SI}$ [6]; When the short axis of the particles is placed parallel on the loading plate, Hiramatsu and Oka [13] proposed that $d_0 = S$; For massive particles, due to their similar dimensions in the three axial directions, Wu et al. [51] proposed that $d_0 = (L+I+S)/3$ or $d_0 = \sqrt{LSI}$. On this basis, Bagheri et al. [3] revised the formula as $d_0 = 0.887\sqrt[3]{LIS}$ based on the rough and porous characteristics of the particle surface. Using this method, results were compared with those obtained from other approaches, showing consistent trends in breakage strength, with less than 5 % difference in characteristic strength and a slight (≈5–8 %) increase in the Weibull modulus, indicating reduced dispersion, and this method can better describe the three-dimensional stress situation and effectively control the average error.

The survival probability under different load of particle aggregate is obviously different due to the difference of individual particle breakage strength. Based on this, Weibull proposed a statistical method for brittle materials namely Weibull distribution function [45]. Based on the empirical formula of Weibull function, the statistical analysis of single particle breakage strength using Weibull distribution function is of certain suitability under the two assumptions that the test particles are similar in shape as blocks and the particles are broken due to internal tension. Therefore, the dispersion degree of particle breakage strength of construction waste can be quantitatively described:

$$P_s = \exp\left[-\left(\frac{\sigma_f}{\sigma_{f0}}\right)^m\right] \tag{2}$$

in which P_s is the ratio of the number of unbroken particles at $\sigma_f \geq \sigma_{f0}$ to the total number of tested particles, which can be called the survival probability of particles; σ_f is the breakage strength at failure; σ_{f0} is the characteristic strength, the value of which can be expressed as the corresponding particle breakage stress when the survival probability is 37 % [25] (P_s=1/e); m is Weibull modulus and it can be calculated by Eq. (3):

$$\ln\left(\ln\frac{1}{P_s}\right) = m\ln\left(\frac{\sigma_f}{\sigma_{f0}}\right) \tag{3}$$

The m can be used to characterize the dispersion or variability of particle breakage strength. A larger difference of m generally indicates a smaller dispersion of the strength distribution, and vice versa.

3. Experimental results and discussion

3.1. Single-particle fractal dimension

According to fractal theory, the essence of single particle breakage is the change of particle size grading curve, and particle size grading curve follows fractal theory [9,49]:

$$F(d) = \left(\frac{d}{d_M}\right)^{3-D} \tag{4}$$

In Eq. (4): $F(d)$ refers to the mass proportion of particles less than a certain size range; d_M is the maximum size of the crushed particles; D is the fractal dimension.

The fractal dimension ($D\in[2,3)$) serves as a quantitative measure of fragmentation complexity, characterizing the power-law relationship between particle size distribution and fragmentation energy dissipation. During the particle breakage process, the trend of self-phase of broken shapes also increases, resulting in a higher fractal dimension.

The fractal dimension of particles generally varies among different breakage tests. After the loading process of the single-particle breakage test, the crushed fragments of RM, RB, and RC were collected, and their particle size distributions were determined through standard sieve analysis. Fig. 4(a)-(c) present the particle size distributions of RM, RB, and RC before and after breakage, respectively. As shown in the figures, particle breakage caused the gradation curves to shift toward the upper left of the coordinate system. Under

equivalent particle size ranges, RB and RM exhibited higher D values than RC in the logarithmic coordinate system, indicating more extensive fragmentation of RM and RB compared with RC, and consequently a higher particle strength for RC. Furthermore, the fractal dimension increased with particle size for all three materials. This trend arises because larger particles tend to contain more internal flaws and experience higher stress concentrations, making them more susceptible to breakage. In addition, the increases in D values of RM and RB, as the particle size range increased from 4.75 to 10 mm to 20–25 mm, were 0.34 and 0.33, respectively—much larger than that of RC (0.21). This phenomenon can be attributed to differences in the breakage modes among various construction waste particles, which will be discussed later.

Fig. 4(d) illustrates the composite particle size distributions for the three materials under identical size fractions. The double-logarithmic gradation curves yielded slopes of 0.66 (RM), 0.74 (RB), and 0.88 (RC). Applying Eq. (4) to the single-particle compression test data, the calculated D values for RM, RB, and RC were 2.34, 2.26, and 2.12, respectively. These results align well with previous studies, which reported D values of approximately 2.34 for construction slag [42], 2.50–2.60 for mixed demolition waste [50], and 2.31–2.40 for recycled aggregates [5]. The measured range of fractal dimensions (2.12–2.34) corresponds well with established fragmentation models for construction waste, with observed variations primarily attributed to differences in particle morphology and loading configurations.

3.2. Particle breakage strength

3.2.1. Effect of particle size on breakage strength

Fig. 5 illustrates the Weibull distributions characterizing the variability in breakage strength of recycled materials across different particle size ranges. The maximum characteristic strength σ_{f0} was observed in the 4.75–10 mm size fraction for all material types. A progressive decrease in σ_{f0} with increasing particle size was evident, and the 20–25 mm particles exhibited the lowest σ_{f0} values, demonstrating pronounced size-dependent behavior consistent with Weibull statistical theory. This phenomenon arises because larger particles generally contain a higher density of internal flaws and experience greater stress concentrations during loading [21,22]. On the one hand, larger particles typically have more irregular morphologies than smaller ones, with numerous pre-existing cracks and pores both on the surface and within the interior, leading to relatively lower strength. Moreover, larger particles require more energy to undergo breakage during loading. Once the applied load exceeds the particle's breakage strength, the release of stored energy

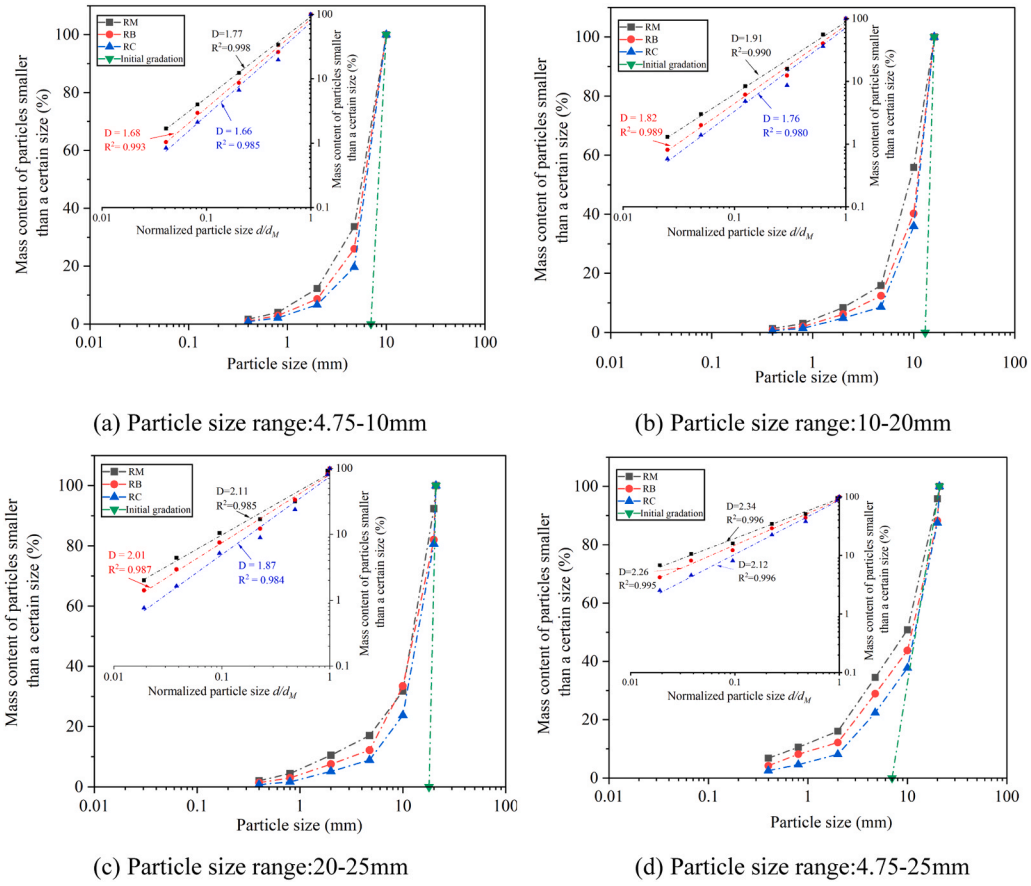
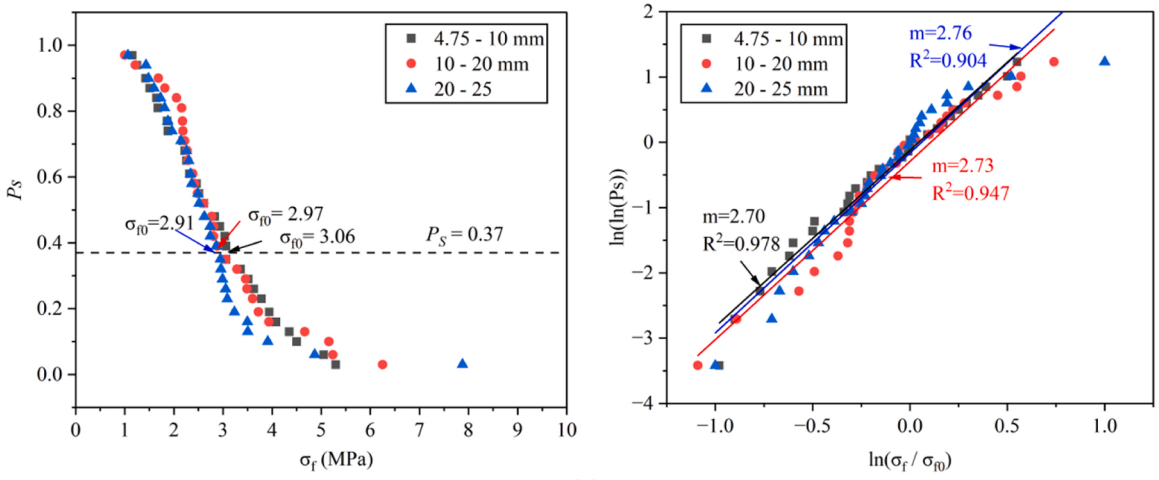
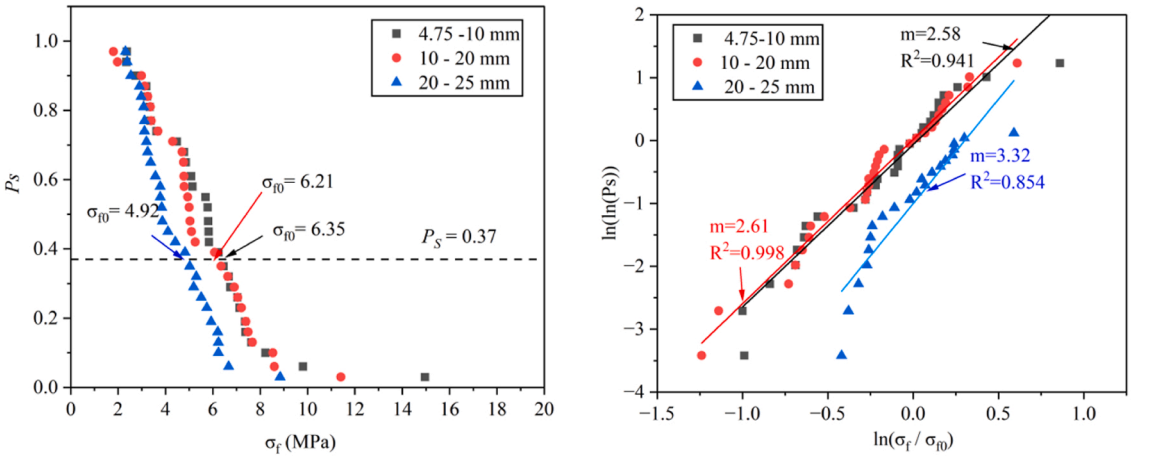


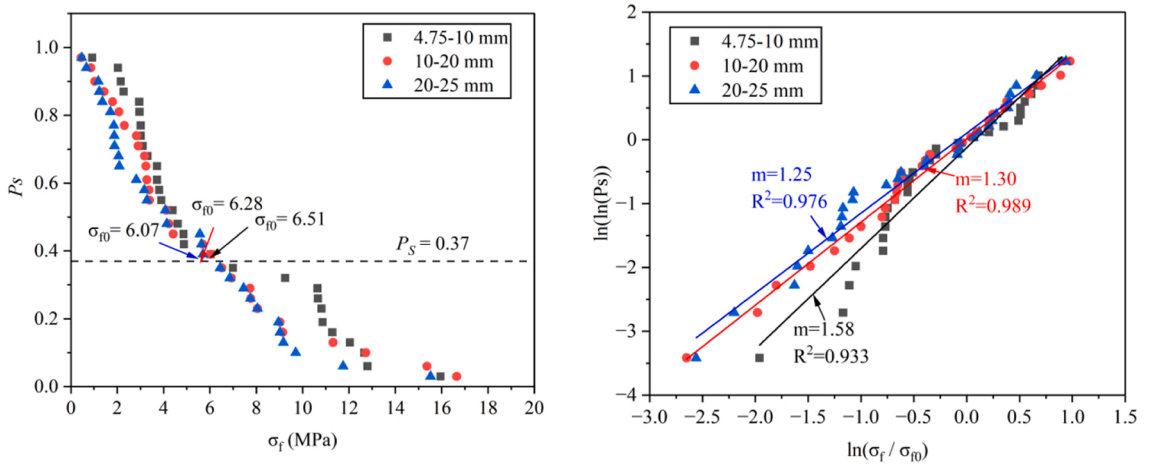
Fig. 4. Fractal dimension of particle breakage of construction waste.



(a) RM



(b) RB



(c) RC

Fig. 5. Survival probability curve and Weibull distribution for different particle materials and size.

becomes more intense, resulting in more extensive fragmentation.

According to the Weibull distributions shown in Fig. 5, the Weibull modulus m increased with particle size for RM and RB, indicating improved strength homogeneity in the 20–25 mm range. In contrast, m for RC decreased with increasing size. This can be attributed to the complex pore networks and interfacial defects within RC particles; as particle size increases, both the number of these flaws and the heterogeneity of their spatial distribution increase, thereby enhancing the variability of breakage strength.

3.2.2. Effect of particle type on breakage strength

Table 2 shows that within the 4.75–25 mm particle size range, the average characteristic strength $\overline{\sigma}_{f0}$ of RM, RB, RC were 2.97 MPa, 5.38 MPa, and 6.28 MPa, respectively. The fractal dimension analysis confirms that RC exhibits the highest $\overline{\sigma}_{f0}$, while RM showed the lowest $\overline{\sigma}_{f0}$, indicating higher susceptibility to breakage. In addition, it can be seen that the average m of RM, RB and RC were 2.73, 2.84 and 1.38, respectively, reflecting material-dependent variations in distribution of breakage strength. RC exhibited significantly greater strength dispersion compared to RM and RB, with this variability becoming more pronounced with increasing particle size. This behavior may originate from the heterogeneous type of RC, containing mortar-cement aggregates with rough surfaces and complex pore structures [30]. Larger RC particles contain abundant, densely packed pores with non-uniform spatial configurations, creating structural heterogeneity within individual particles. Under compressive loading, preferential failure occurs at weakened microstructural regions, while less defective zones maintain load-bearing capacity, amplifying strength variability. On the contrary, pore structures of RB particles develop increasing uniformity at larger sizes [30], enhancing structural stability and reducing strength dispersion, Consequently, RB exhibited the highest average m value.

3.3. Particle breakage mode

Single-particle breakage patterns are governed by material type, morphological characteristics, and particle size. Cai et al. [5] categorized compressed recycled aggregates particles of different shapes into penetrating fracture, corner rupture, and overall fragmentation. Tang et al. [40] identified four distinct failure mechanisms in limestone particulates through coupled mechanical analysis: explosive fragmentation, axial splitting, progressive breakage, and localized edge failure. Generally, these breakage modes were categorized based on criteria such as the post-breakage particle size and the cracking pattern. However, the breakage process of construction waste typically involves multiple concurrent failure mechanisms. Construction waste particles exhibit phase-dependent and load-bearing characteristics during fragmentation, with failure mode transitions corresponding to specific energy dissipation thresholds, but little attention has been paid to the relationship between particle breakage behavior and the force-displacement response, which has received limited attention in previous studies. In this study, combining parameters such as the fractal model of the crushed particles, the peak load value, the force-displacement curve (corresponding to the crack propagation of the particles), and the range of fragment numbers, the differentiated breakage mechanisms of RC, RB, and RM were analyzed, and the corresponding breakage modes under the combined effects of particle size and material type.

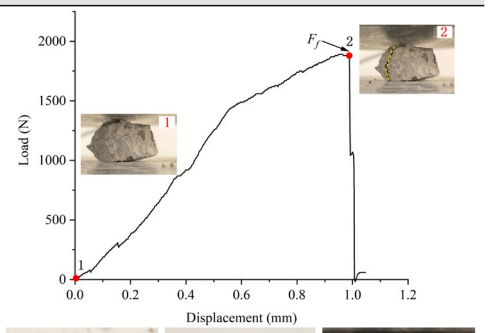
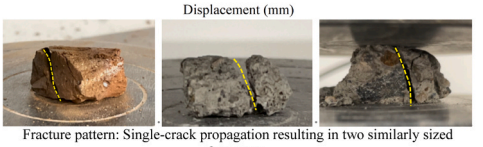
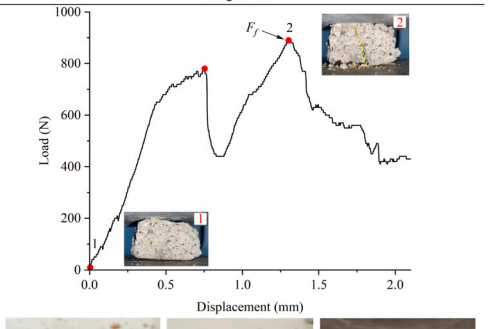
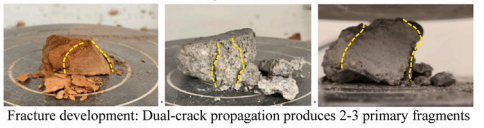
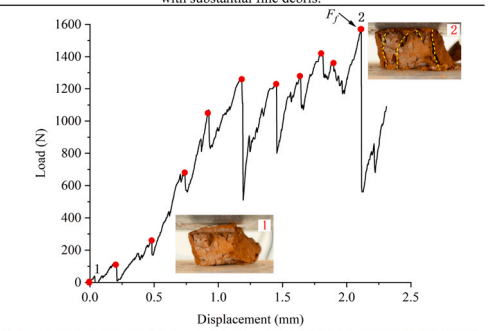
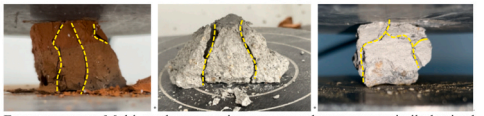
Based on the analysis of 270 single-particle compression tests, three parallel tests indicated good repeatability of all particle breakage modes, with standard deviations below 10 % and short error bars, demonstrating the good stability of the experimental data. Three dominant breakage modes for construction waste particulates were identified: intermediate splitting (Mode I), mixed breakage (Mode II), and bulk breakage (Mode III). Table 3 summarizes the detailed morphological and mechanical characteristics for each breakage mode：

- Intermediate splitting (Mode I): In the initial stage, the load increases with the increase of displacement linearly, and then decreases sharply with particle breakage. The force-displacement curve is relatively smooth with a few jagged and convex points. Generally, only one single peak point can be seen on the curve. At the corresponding peak point, the broken particles directly split into 2 pieces, accompanied by a small number of debris. the number of fragments is generally no more than two ($\leq$2). After breaking, one vertical crack is observed.
- Mixed breakage (Mode II): The load increases with the displacement linearly at the initial stage and then decreases sharply when the first crack is generated on the particle and the force-displacement curve reaches the first peak point. Different from the first mode, the curve rises again as the displacement continuously increases and reaches the second peak point. Finally, the curve declines gradually after breaking. There are two load peak points on the curve, and at the corresponding peak points, the particle

Table 2
Test results.

Material group		Particle size (mm)	Characteristic strength (σ_{f0}/MPa)	Average characteristics strength ($\overline{\sigma}_{f0}$/MPa)	Weibull modulus (m)	Average Weibull modulus ($\overline{m}$)
RM	RM-1	4.75–10	3.03	2.97	2.70	2.73
	RM-2	10–20	2.97		2.73	
	RM-3	20–25	2.91		2.76	
RB	RB-1	4.75–10	6.35	5.83	2.58	2.84
	RB-2	10–20	6.21		2.61	
	RB-3	20–25	4.92		3.32	
RC	RC-1	4.75–10	6.51	6.28	1.58	1.38
	RC-2	10–20	6.28		1.30	
	RC-3	20–25	6.07		1.25	

Table 3
The particle breakage mode of construction waste.

Breakage mode	Particle characteristics
Intermediate splitting	  Fracture pattern: Single-crack propagation resulting in two similarly sized fragments.
Mixed breakage	  Fracture development: Dual-crack propagation produces 2-3 primary fragments with substantial fine debris.
Bulk breakage	  Fracture pattern: Multi-crack propagation generates three or more similarly sized fragments with minor debris.

splits into 2–3 fragments. With a few cracks, contact fractures were observed on the particle surface, resulting in a large number of sharp-angled small fragments.

- Bulk breakage (Mode III): The load increases with the displacement gradually with multiple peak points on the curve. In contrast with the first two modes, no linearly increasing stage can be observed on the curve. Particle breakage generates multiple cracks and produces three or more fragments of comparable size ($\geq$3), Multiple cracks with at least one main crack and a few debris are observed.

Fig. 6 presents the proportions of the three breakage modes for different types of construction waste particles. According to Fig. 6 (a), bulk breakage and mixed breakage are the predominant breakage modes for RM particles, while the proportion of intermediate splitting is only 13 %. However, the predominant breakage modes of RM are strongly correlated with particle size. For small RM particles (d_0<20 mm), over 50 % of the particles undergo bulk breakage. As particle size increases, the proportion of mixed breakage also rises, making it the predominant breakage mode for RM particles with $d_0\geq$ 20 mm. The breakage modes of RB are similar to those of RM, and the predominant breakage mode is likewise correlated with particle size. The proportions of mixed and bulk breakage mode are much higher than that of intermediate splitting. Bulk breakage mode is predominant in particles with $d_0<$ 20 mm, whereas mixed breakage predominates in large particles with $d_0\geq$ 20 mm. On the other hand, the breakage mode of RC particles shows a distinctly different distribution. For small RC particles with d_0 range from 4.75 mm to 10 mm, the proportions of the three breakage modes are nearly equal. As d_0 increases, intermediate splitting becomes the predominant breakage mode. This phenomenon can be largely attributed to the graded composite nature of RC aggregates. Coarse RC particles ($d_0\geq$10 mm) generally consist of a gravel core surrounded by mortar. During single-particle compression, failure initiates with surface mortar spallation, followed by vertical crack propagation through the gravel core, ultimately resulting in a median split at the critical stress threshold. In contrast, smaller RC particles (4.75–10 mm) consist primarily of mortar mixed with several small gravel fragments. The complex natural flaws at the interfaces between different types facilitate rapid crack propagation, producing double or multiple cracks and corresponding double- or multi-peak force-displacement responses, as well as 3–5 major fragments. Since mixed and bulk breakage modes generate more small fragments, the D values of RM and RB are larger than those of RC, as shown in Fig. 4.

Combining the test results as presented in Fig. 6 and Table 2, it seems that the average characteristic strength $\overline{\sigma}_{f0}$ is correlated with the breakage mode. In the intermediate splitting mode, the particle maintains integrity before breakage, leading to a relatively high

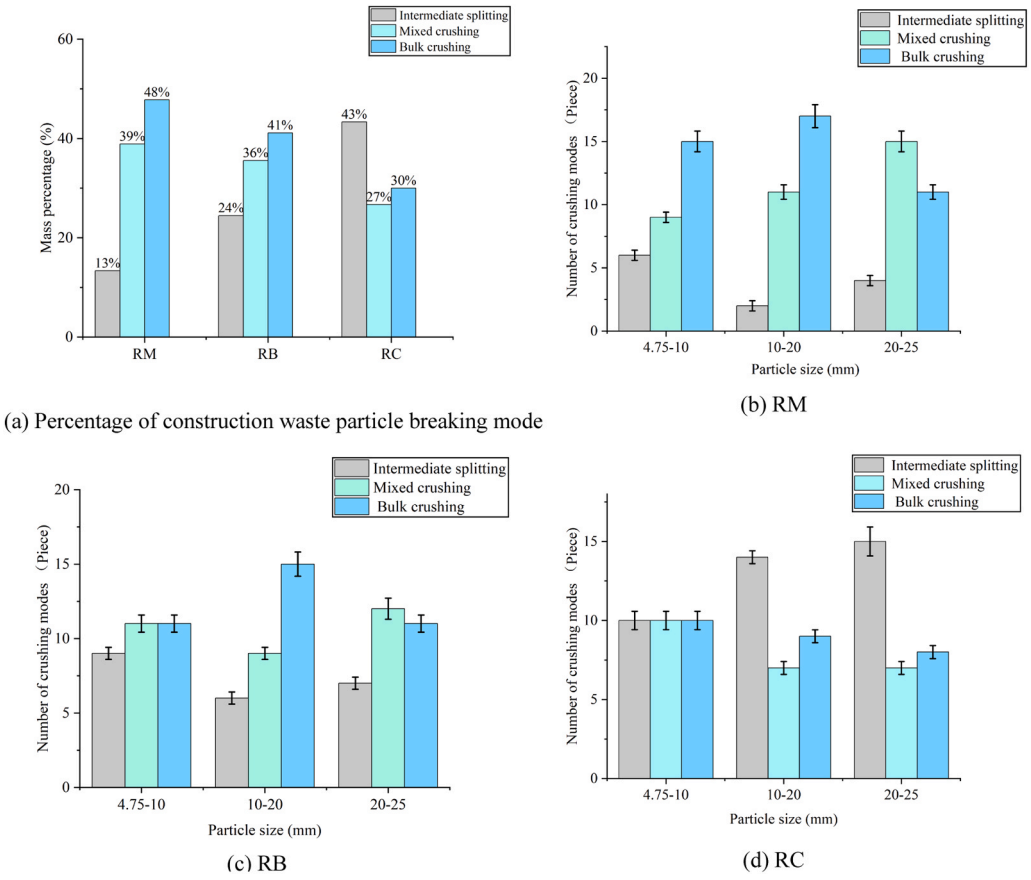


Fig. 6. Statistics of particle breakage mode of construction waste.

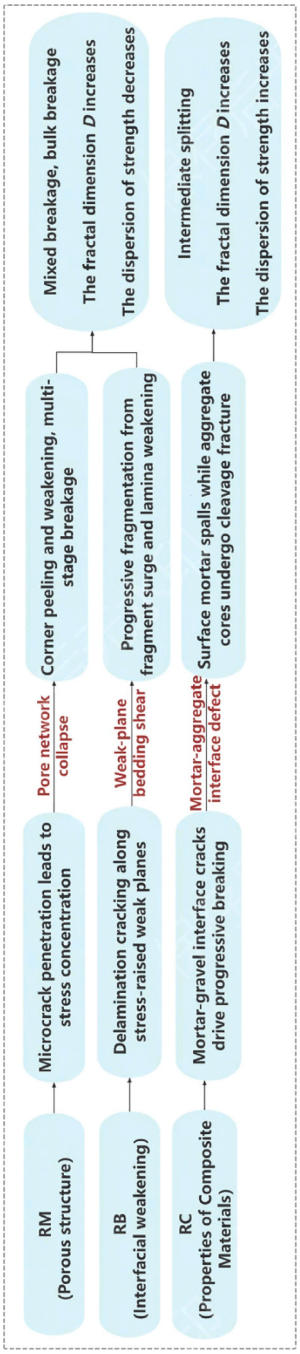


Fig. 7. Diagram of the breakage mechanisms of different types.

breakage strength. However, cracks were generated inside the particles before the failure point in the mixed breakage mode and the bulk breakage mode, which will reduce the particle breakage strength. As the RC particles breakage is dominated by the intermediate splitting mode, $\overline{\sigma}_{f0}$ of RC particles is larger than that of RM and RB particles, and $\overline{\sigma}_{f0}$ of RM particles is smallest among the three kinds of construction waste particles because little particles breakage accord with the intermediate splitting mode. According to the analysis above, the key mechanisms of particle breakage under the coupling effect of types and particle size of construction waste are summarized in Fig. 7.

4. Discussion

This study revealed that particle size and material type jointly govern the breakage behavior of construction waste. The coupling of these two factors significantly influences the breakage strength, fractal characteristics, and failure modes of recycled materials. For RC particles with high mortar content, smaller particles predominantly undergo bulk breakage, whereas larger ones mainly exhibit intermediate splitting. This indicates that optimizing input energy and size gradation can reduce over-pulverization and improve yield efficiency. In contrast, RM and RB particles display relatively stable strength-size relationships, suggesting that single-stage crushing followed by sieving is sufficient to control particle size and avoid redundant energy consumption. These findings provide practical guidance for regulating breakage energy, controlling target particle-size distributions, and improving the efficiency and uniformity of recycled aggregate production.

From a material design perspective, distinct breakage patterns and strength distributions among RC, RM, and RB particles reflect the decisive role of microstructural features in governing macroscopic mechanical behavior. By tailoring mixture proportions and gradation based on each type's breakage characteristics, recycled concrete and subgrade materials can achieve an optimal balance between strength and workability. In particular, enhancing interfacial bonding and controlling coarse aggregate content in RC-based materials can reduce splitting failures and improve load-bearing stability.

The present research is limited to single-particle compression tests and does not account for inter-particle interactions, moisture effects, or internal defect distributions. Future work will employ CT-based and XRD/XRF quantitative characterization, multi-particle contact tests, and numerical simulations to establish the relationships among breakage energy, fragment-size distribution, and fractal dimension, incorporate 3D image reconstruction and quantitative morphology parameters (e.g., sphericity, aspect ratio) to further evaluate the evolution of particle shape before and after breakage thereby improving the understanding and predictive modeling of recycled aggregate breakage mechanisms.

5. Conclusion

This study systematically investigated the breakage behaviors of three kinds of construction waste particles: recycled mortar (RM), recycled brick (RB), and recycled concrete (RC), through single-particle compression experiments across varied size fractions (4.75–25 mm). The Weibull statistical framework was employed to quantify breakage strength characteristics and strength distribution variability, while fractal dimension analysis of pre- and post-fragmentation gradation curves provided comminution intensity quantification. According to the test results, three breakage modes are proposed to describe the breakage behaviors of different construction wastes and the coupled effect of particle size and type is carefully analyzed. The principal findings demonstrate:

(1) The particle size and type jointly exert a coupled effect on the single-particle breakage strength of construction waste. The average characteristic strength $\overline{\sigma}_{f0}$ of RC particles is approximately twice that of RM particles, representing the highest and lowest values, respectively, among the three types of materials. Furthermore, the$\overline{\sigma}_{f0}$of all particle types decreases progressively with increasing particle size, consistent with the Weibull statistical distribution. The average Weibull modulus $\overline{m}$ increases with particle size for RM and RB particles, suggesting enhanced strength uniformity. In contrast, RC particles exhibit a decreasing $\overline{m}$ value with increasing particle size, implying that the structural heterogeneity of RC particles becomes more pronounced as the particle size increases.

(2) Particle size and type jointly influence the fractal characteristics of construction waste particles. At the same particle size, the average fractal dimension D of RM and RB particles is approximately 10 % higher than that of RC, indicating a higher degree of breakage. Considering that RC particles exhibit higher characteristic strength, it can be inferred that RM and RB particles are more prone to fragmentation under equivalent loading conditions. Moreover, D increases with particle size for all particle types, with the increments in RM and RB being approximately 50 % greater than that of RC, reflecting distinct breakage mechanisms among different types of construction waste particles.

(3) Based on the recorded breakage sequences and load–displacement curves, three typical breakage modes of construction waste particles were identified: intermediate splitting, mixed breakage, and bulk breakage. Both particle size and type jointly influence the dominant breakage mode. For RM and RB particles, bulk breakage is predominant when the particle size is small ($d_0 \leq 20$ mm). With increasing particle size, the proportion of mixed breakage gradually increases and surpasses that of bulk breakage when $d_0 > 20$ mm. In contrast, for RC particles, the three breakage modes occur in approximately equal proportions at small sizes ($d_0 \leq 10$ mm), whereas the intermediate splitting mode becomes dominant as particle size increases.

CRediT authorship contribution statement

Changjie Xu: Supervision. **Xiaozhen Fan:** Conceptualization. **Yujie Shi:** Investigation. **Jinshan Tong:** Writing – original draft,

Validation, Formal analysis, Data curation. **Luju Liang:** Writing – review & editing, Resources, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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